DO NOW — RECALL

Hashing and applications

Name: _____ Class: _____ Date: _____

1. What is the encryption key in symmetric encryption?

2. What is the role of a public key?

3. Why encrypt a message in transit?

4. Compute 10110110_2 AND 00001111_2, then hexadecimal.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A shared secret used to encrypt and decrypt.

2. Model answer

Can be distributed for encryption or verification according to the scheme.

3. Model answer

To protect its content from unintended readers.

4. Model answer

$00000110_2 = 06_{16}$; high four bits masked away.

Vocabulary — know these terms

hash — Fixed-size digest of arbitrary input.

collision — Two inputs producing the same digest.

salt — Random value mixed into password hashing.

KDF — Key/password derivation function.

integrity — Confidence that data has not changed.

NEW LEARNING

Read, annotate and explain

Understand

A cryptographic hash transforms an arbitrary input into a fixed-size digest. It is designed to make reversing the digest infeasible, not impossible under all circumstances.

Trace

A message digest supports integrity checking when the reference hash is trusted. Hash collisions can exist because outputs have finite length.

Evaluate

Passwords should be processed using a salted, purpose-built slow password-hashing scheme, not simply a fast general-purpose hash and never reversible encryption.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A downloaded file's trusted SHA-256 checksum differs from the computed checksum. What can you conclude?

Predict first

The file contents or expected reference differ: the integrity check fails.

Show method

The mismatch does not by itself reveal whether alteration was accidental or malicious.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Define cryptographic hashing and distinguish it from reversible encryption. [6]

Question 2

Explain how a trusted checksum helps verify a software download and give one limitation. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain salted password hashing and why storing raw passwords or just fast unsalted hashes is weaker. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

If a password is hashed, the server simply decrypts that hash during login.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.